Palo Alto Networks PAN-OS — Advanced Routing & Configuration Master Guide

Complete 72-Topic Engineering Field Manual: Detailed Technical Meanings, Concrete Real-World Examples, PAN-OS CLI Syntax, and Web GUI Step-by-Step Paths across IPv4/IPv6 Static, RIP, OSPF, BGP, Path Monitoring, and Multi-Area Topologies

Document ID:	PAN-ENG-ROUTING-72	**Platform Support:**	PA-220 / 400 / 800 / 3200 / 5200 / PA-VM
Target OS Train:	PAN-OS 9.1 / 10.x / 11.x (Virtual Router)	**Routing Protocols:**	Static, RIPv2, OSPFv2, BGPv4, IPv6 OSPFv3, MP-BGP
Scope:	Enterprise Production & PCNSE Certification	**Lead Engineer:**	Nishu Saini (Innspark Solutions)

Module 1: Initial Setup, Management Plane & Security Zone Architecture

Topic 1: Initial Configuration: Basics via GUI & CLI

■ **Meaning & Architectural Concept:** Establishing the out-of-band management IP, netmask, default gateway, DNS resolvers, and device hostname. This binds the dedicated MGT interface on the control plane, separating firewall management traffic from data plane user flows.

■ ***Concrete Real-World Example:*** *Staging a fresh PA-VM in a lab environment with IP 172.17.0.50/16, gateway 172.17.0.1, primary DNS 8.8.8.8, secondary DNS 8.8.4.4, and hostname PA-VM-LAB.*

```
admin@PA-VM> configure
admin@PA-VM# set deviceconfig system type static
admin@PA-VM# set deviceconfig system ip-address 172.17.0.50 netmask 255.255.0.0 default-gateway 172.17.0.1
admin@PA-VM# set deviceconfig system dns-setting servers primary 8.8.8.8 secondary 8.8.4.4
admin@PA-VM# set deviceconfig system hostname PA-VM-LAB
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. Connect browser to **https://172.17.0.50** and log in.
2. Navigate to: **Device** → **Setup** → **Management**.
3. Click **Gear (■■)** on *Management Interface Settings* → Type: Static, IP: 172.17.0.50, Netmask: 255.255.0.0, Gateway: 172.17.0.1 → Click **OK**.
4. Click **Gear (■■)** on *General Settings* → Hostname: PA-VM-LAB → **OK** → **Commit**.

Topic 2: Initial Configuration: Security Zone Implementation

■ **Meaning & Architectural Concept:** Palo Alto firewalls enforce a Zero-Trust architecture where all interfaces must belong to a Security Zone. Traffic between different zones is blocked by default until an explicit Security Policy rule allows it; traffic within the same zone is allowed by default.

■ ***Concrete Real-World Example:*** *Creating four network zones: INSIDE (Trust LAN), DMZ1 (Web Servers), OUTSIDE (Untrust WAN), and DMZ2 (Database Servers), binding them to physical ports ethernet1/1 through ethernet1/4.*

```
admin@PA-VM# set zone INSIDE network layer3 ethernet1/1
admin@PA-VM# set zone DMZ1 network layer3 ethernet1/2
admin@PA-VM# set zone OUTSIDE network layer3 ethernet1/3
admin@PA-VM# set zone DMZ2 network layer3 ethernet1/4
admin@PA-VM# commit
admin@PA-VM> show zone
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Zones** → Click **Add**.
2. Name: INSIDE → Type: Layer3 → Under *Interfaces*, click **Add** → Select ethernet1/1 → Click **OK**.
3. Repeat for DMZ1 (ethernet1/2), OUTSIDE (ethernet1/3), and DMZ2 (ethernet1/4) → Click **Commit**.

Module 2: IPv4 Static Routing, Default Gateway & Path Monitoring

Topic 3: Interface Configuration: IPv4 Static Routing GUI/CLI

■ **Meaning & Architectural Concept:** Manually programming destination network prefixes into the Virtual Router table pointing to a specific next-hop IP address or outgoing interface. Static routes have an administrative distance (metric) of 10 by default.

■ ***Concrete Real-World Example:*** *Directing corporate user traffic destined for branch server pool 10.20.0.0/16 to the DMZ1 router next-hop 192.168.2.1 via interface ethernet1/2.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route Route-DMZ1-Pool destination 10.20.0.0/16 interface ethernet1
admin@PA-VM# commit
admin@PA-VM> show routing route virtual-router default
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Virtual Routers** → Click **default**.
2. Select **Static Routes** → **IPv4** → Click **Add**.
3. Name: Route-DMZ1-Pool → Destination: 10.20.0.0/16 → Interface: ethernet1/2 → Next Hop: IP Address → Value: 192.168.2.1 → Metric: 10 → **OK** → **Commit**.

Topic 4: Interface Configuration: IPv4 Default Routing GUI/CLI

■ **Meaning & Architectural Concept:** The gateway of last resort (quad-zero 0.0.0.0/0). Any packet whose destination IP does not match any specific route in the routing table is forwarded out this default path to the upstream ISP router.

■ ***Concrete Real-World Example:*** *Routing all unmapped public Internet traffic out interface ethernet1/3 to the primary ISP border gateway at 192.168.3.1.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route Default-Route destination 0.0.0.0/0 interface ethernet1/3 ne
admin@PA-VM# commit
admin@PA-VM> show routing fib virtual-router default
```

Web GUI Step-by-Step Navigation:
1. In **Network** → **Virtual Routers** → default → **Static Routes** → **IPv4**.
2. Click **Add** → Name: Default-Route → Destination: 0.0.0.0/0 → Interface: ethernet1/3 → Next Hop IP: 192.168.3.1 → **OK** → **Commit**.

Topic 5: Interface Configuration: IPv4 Path Monitoring GUI/CLI

■ **Meaning & Architectural Concept:** Attaching continuous ICMP keepalive health probes to a static route. If the monitored upstream target stops responding, PAN-OS invalidates the static route and pulls it from the active Forwarding Information Base (FIB), enabling automated dual-ISP failover.

■ ***Concrete Real-World Example:*** *Probing upstream DNS 8.8.8.8 sourced from ethernet1/3 (192.168.3.2) every 3 seconds. If 5 consecutive pings fail, withdraw the primary ISP default route so backup ISP route (metric 20) takes over.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route ISP1-Default destination 0.0.0.0/0 interface ethernet1/3 nex
admin@PA-VM# set network virtual-router default routing-table ip static-route ISP1-Default path-monitor enable yes failure-condition any
admin@PA-VM# commit
admin@PA-VM> show routing path-monitor
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **Static Routes** → Edit ISP1-Default.
2. Select **Path Monitor** tab → Check **Enable** → Condition: Any.
3. Click **Add** → Name: Mon-DNS → Dest: 8.8.8.8, Source: 192.168.3.2, Interval: 3, Count: 5 → **OK** → **Commit**.

Module 3: Advanced Routing — IPv4 RIP (Routing Information Protocol)

Topic 6: Advance-Routing: IPv4 RIP GUI/CLI

■ **Meaning & Architectural Concept:** Enabling the distance-vector Routing Information Protocol (RIPv2) on the virtual router. RIP uses hop count as its metric (maximum 15 hops; 16 represents infinity/unreachable).

■ ***Concrete Real-World Example:*** *Activating RIP routing engine on virtual router default to interoperate with legacy edge switches.*

```
admin@PA-VM# set network virtual-router default rip enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Virtual Routers** → Click **default**.
2. Select the **RIP** tab → Check the **Enable** box → Click **OK** → **Commit**.

Topic 7: Advance-Routing: IPv4 RIP Lab-Basic

■ **Meaning & Architectural Concept:** Configuring RIP Interface Groups to determine which firewall interfaces broadcast and listen for RIPv2 multicast updates on 224.0.0.9 (UDP port 520).

■ ***Concrete Real-World Example:*** *Binding interface ethernet1/1 (peer to Router-1) and ethernet1/2 (peer to Router-2) to RIP-Group1 in normal mode.*

```
admin@PA-VM# set network virtual-router default rip interface-groups RIP-Group1 interface ethernet1/1 mode normal
admin@PA-VM# set network virtual-router default rip interface-groups RIP-Group1 interface ethernet1/2 mode normal
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **RIP** → **Interface Groups**.
2. Click **Add** → Name: RIP-Group1 → Under *Interfaces*, add ethernet1/1 and ethernet1/2 (Mode: normal) → **OK** → **Commit**.

Topic 8: Advance-Routing: IPv4 RIP Lab-Routing Table Verification

■ **Meaning & Architectural Concept:** Auditing the Routing Information Base (RIB) to verify which dynamic prefixes have been learned from adjacent RIP routers, along with their hop counts and next-hop IPs.

■ ***Concrete Real-World Example:*** *Confirming that remote branch subnet 10.10.1.0/24 is learned via RIP from neighbor 192.168.1.1 with a hop count of 2.*

```
admin@PA-VM> show routing protocol rip route
admin@PA-VM> show routing route protocol rip virtual-router default
```

Web GUI Step-by-Step Navigation:
1. Go to: **Network** → **Virtual Routers** → Click **More Runtime Stats** on default.
2. Select **Routing Table** tab → Filter flag **R** (RIP) to observe prefix, next-hop, and metric.

Topic 9: Advance-Routing: IPv4 RIP Lab-Peer Table Verification

■ **Meaning & Architectural Concept:** Validating active RIP neighbors, verifying peer reachability, last update received timestamp, and detecting silent or timed-out routers.

■ ***Concrete Real-World Example:*** *Checking neighbor table to verify peer 192.168.1.1 has communicated within the last 30 seconds.*

```
admin@PA-VM> show routing protocol rip peer
admin@PA-VM> show routing protocol rip interface
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → Select **RIP** tab.
2. Click **Peers** to inspect neighbor IP, status, and last update received timestamp.

Topic 10: Advance-Routing: IPv4 RIP Lab-Redistribution

■ **Meaning & Architectural Concept:** Injecting non-RIP routes (such as directly connected LAN subnets or static routes) into RIP update advertisements so neighboring routers discover them.

■ ***Concrete Real-World Example:*** *Redistributing local connected subnets into RIP with metric 2 so Router-1 and Router-2 learn firewall local LAN networks.*

```
admin@PA-VM# set network virtual-router default rip-redist-rules Redist-Connected enable yes metric 2 route-table unicast filter match-rou
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **RIP** → **Redistribution Rules**.
2. Click **Add** → Name: Redist-Connected → Metric: 2 → Filter: Check **connect** → **OK** → **Commit**.

Topic 11: Advance-Routing: IPv4 RIP Lab-Timer Modification

■ **Meaning & Architectural Concept:** Tuning RIP update (default 30s), route timeout (default 180s), and garbage collection (default 120s) timers. Modifying these timers accelerates network convergence after link failures.

■ ***Concrete Real-World Example:*** *Adjusting RIP update timer to 15s, timeout to 90s, and garbage collection to 60s for 2x faster failover in a high-speed LAN.*

```
admin@PA-VM# set network virtual-router default rip timers update 15 timeout 90 garbage-collection 60
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **RIP** → **Timers**.
2. Set Update: 15, Timeout: 90, Garbage Collection: 60 → **OK** → **Commit**.

Topic 12: Advance-Routing: IPv4 RIP Lab-Advertise Default-Route

■ **Meaning & Architectural Concept:** Instructing the firewall to generate and advertise a default route (0.0.0.0/0) in its RIP updates, pointing downstream routers toward the firewall for all external traffic.

■ ***Concrete Real-World Example:*** *Advertising default route with metric 1 to all RIP peers so branch routers send all Internet-bound packets to the firewall.*

```
admin@PA-VM# set network virtual-router default rip default-route-advertise enable yes metric 1
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **RIP**.
2. Under *Default Route* → Check **Advertise** → Metric: 1 → **OK** → **Commit**.

Topic 13: Advance-Routing: IPv4 RIP Lab-Authentication

■ **Meaning & Architectural Concept:** Securing RIPv2 packet updates with MD5 cryptographic authentication to prevent unauthorized routers from spoofing routing updates or poisoning the routing table.

■ ***Concrete Real-World Example:*** *Configuring MD5 authentication with Key ID 1 and shared key SecretKey123 on interface group RIP-Group1.*

```
admin@PA-VM# set network virtual-router default rip interface-groups RIP-Group1 authentication md5 key-id 1 key SecretKey123
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **RIP** → Click RIP-Group1.
2. Authentication: Select MD5 → Key ID: 1 → Key: SecretKey123 → **OK** → **Commit**.

Topic 14: Advance-Routing: IPv4 RIP Troubleshoot

■ **Meaning & Architectural Concept:** Diagnosing RIP protocol issues, including count-to-infinity routing loops, split-horizon hold-downs, authentication mismatches, and firewall security policy drops on UDP 520.

■ ***Concrete Real-World Example:*** *Enabling routing engine debug tracing and tailing routed.log to detect rejected unauthenticated packets from neighbor 192.168.1.1.*

```
admin@PA-VM> show routing protocol rip status
admin@PA-VM> debug routing pfe rip all on
admin@PA-VM> tail follow yes mp-log routed.log
```

Web GUI Step-by-Step Navigation:
1. Go to: **Monitor** → **Logs** → **System** → Filter: (subtype eq routing) and (description contains 'rip').
2. Inspect authentication mismatches, metric 16 unreachable errors, and packet drop events.

Module 4: Enterprise IPv4 OSPF (Open Shortest Path First) Architecture

Topic 15: Advance-Routing: IPv4 OSPF GUI/CLI

■ **Meaning & Architectural Concept:** Enabling the link-state OSPFv2 protocol engine on the virtual router and assigning a unique 32-bit dotted-decimal Router-ID to identify the firewall within the autonomous system.

■ ***Concrete Real-World Example:*** *Enabling OSPF on virtual router default with Router-ID 1.1.1.1.*

```
admin@PA-VM# set network virtual-router default protocol ospf enable yes router-id 1.1.1.1
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Virtual Routers** → Click **default** → **OSPF** tab.
2. Check **Enable** → Router ID: 1.1.1.1 → **OK** → **Commit**.

Topic 16: Advance-Routing: IPv4 OSPF Lab-Basic and Multi-Area

■ **Meaning & Architectural Concept:** Configuring a hierarchical multi-area OSPF design. Area 0 (Backbone) connects to core distribution switches, while non-backbone areas isolate flooding domains and reduce SPF recalculation overhead.

■ ***Concrete Real-World Example:*** *Binding ethernet1/1 to backbone Area 0.0.0.0 and ethernet1/2 to branch Area 0.0.0.1.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.0 type normal interface ethernet1/1 enable yes
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.1 type normal interface ethernet1/2 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → **Areas**.
2. Add Area 0.0.0.0 (Type: normal, Interface: ethernet1/1).
3. Add Area 0.0.0.1 (Type: normal, Interface: ethernet1/2) → **Commit**.

Topic 17: Advance-Routing: IPv4 OSPF Lab-Neighbor Table Verification

■ **Meaning & Architectural Concept:** Auditing OSPF neighbor adjacency states. On broadcast multi-access links, neighbors must transition through Down, Init, 2-Way, ExStart, Exchange, Loading, to Full (DR, BDR, or DROther).

■ ***Concrete Real-World Example:*** *Verifying neighbor 192.168.1.1 on ethernet1/1 is in Full state as Designated Router (DR).*

```
admin@PA-VM> show routing protocol ospf neighbor
admin@PA-VM> show routing protocol ospf neighbor detail
```

Web GUI Step-by-Step Navigation:
1. Go to: **Virtual Routers** → **More Runtime Stats** → **OSPF** → **Neighbors**.
2. Confirm neighbor status shows **Full** with DR/BDR role clearly indicated.

Topic 18: Advance-Routing: IPv4 OSPF Lab-Database Table Verification

■ **Meaning & Architectural Concept:** Auditing the Link-State Database (LSDB). All routers in an area must have an identical LSDB containing Router LSAs (Type 1), Network LSAs (Type 2), Summary LSAs (Type 3), and External LSAs (Type 5).

■ ***Concrete Real-World Example:*** *Inspecting the LSDB to ensure all router and network LSAs for Area 0 are synchronized with Router-1.*

```
admin@PA-VM> show routing protocol ospf dump
admin@PA-VM> show routing protocol ospf lsdb
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **OSPF** → **Link State Database**.
2. Inspect LSA counts by type (Router, Network, Summary, AS-External).

Topic 19: Advance-Routing: IPv4 OSPF Lab-Routing Table Verification

■ **Meaning & Architectural Concept:** Verifying OSPF routes installed into the firewall active routing table (FIB) after Dijkstra algorithm computation. Marked with flags: O (Intra-area), Oo (Inter-area), or O1/O2 (External).

■ ***Concrete Real-World Example:*** *Confirming intra-area route 10.10.1.0/24 is installed with next-hop 192.168.1.1 and metric 11.*

```
admin@PA-VM> show routing protocol ospf route
admin@PA-VM> show routing route protocol ospf virtual-router default
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **Routing Table**.
2. Filter by flags **O**, **Oo**, **O1**, **O2** to inspect installed prefixes and metrics.

Topic 20: Advance-Routing: IPv4 OSPF Lab-Virtual-Links

■ **Meaning & Architectural Concept:** Creating a logical link through a non-backbone transit area to reconnect a partitioned area back to Area 0. All OSPF non-backbone areas must connect directly to Area 0; virtual links solve non-contiguous topologies.

■ ***Concrete Real-World Example:*** *Connecting isolated Area 2 through transit Area 1 to reach Area 0, targeting neighbor router ID 3.3.3.3.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.1 virtual-link VL-to-R3 enable yes neighbor-id 3.3.3.3 transit-ar
admin@PA-VM# commit
admin@PA-VM> show routing protocol ospf virtual-link
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Click transit area 0.0.0.1.
2. Select **Virtual Link** tab → Add VL-to-R3 → Neighbor: 3.3.3.3, Transit: 0.0.0.1 → **OK** → **Commit**.

Topic 21: Advance-Routing: IPv4 OSPF Lab-Hello Interval Modification

■ **Meaning & Architectural Concept:** Changing the frequency of OSPF Hello packets sent on an interface (default 10s on broadcast links). Both neighbor routers must have identical Hello intervals to form an adjacency.

■ ***Concrete Real-World Example:*** *Lowering Hello interval on ethernet1/1 to 5s for rapid link-down detection in a datacenter.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/1 hello-interval 5
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Area 0.0.0.0 → Click ethernet1/1.
2. Set **Hello Interval**: 5 → **OK** → **Commit**.

Topic 22: Advance-Routing: IPv4 OSPF Lab-Dead Interval Modification

■ **Meaning & Architectural Concept:** The time a router waits without receiving a Hello packet before declaring the neighbor dead (default 40s, typically 4x Hello). Must match across peers.

■ ***Concrete Real-World Example:*** *Setting Dead interval to 20s to match the 5s Hello interval on ethernet1/1.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/1 dead-counts 20
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Area 0.0.0.0 → Click ethernet1/1.
2. Set **Dead Interval**: 20 → **OK** → **Commit**.

Topic 23: Advance-Routing: IPv4 OSPF Lab-Authentication

■ **Meaning & Architectural Concept:** Encrypting OSPF packet exchanges using MD5 message digests. Routers verify key-id and password hash on every packet, blocking rogue routers from forming adjacencies.

■ ***Concrete Real-World Example:*** *Configuring MD5 authentication with Key ID 1 and password PaloAltoOspfPass1! on interface ethernet1/1 in Area 0.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/1 authentication md5 key-id 1 key PaloAltoO
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Area 0.0.0.0 → Click ethernet1/1.
2. Authentication: MD5 → Key ID: 1 → Key: PaloAltoOspfPass1! → **OK** → **Commit**.

Topic 24: Advance-Routing: IPv4 OSPF Lab-Stub Area

■ **Meaning & Architectural Concept:** Configuring an area to block external Type 4 and Type 5 LSAs. The ABR automatically generates a default route (Type 3 summary) into the stub area, dramatically reducing routing table size.

■ ***Concrete Real-World Example:*** *Configuring branch Area 0.0.0.2 as a Stub Area with default route metric 10 to protect memory on branch firewalls.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.2 type stub default-route metric 10
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → **Areas** → Add Area 0.0.0.2.
2. Type: stub → Default Route Metric: 10 → **OK** → **Commit**.

Topic 25: Advance-Routing: IPv4 OSPF Lab-Totally Stub Area

■ **Meaning & Architectural Concept:** Extending Stub areas by blocking both External LSAs (Type 4/5) AND Inter-Area Summary LSAs (Type 3). The ABR injects only a single default route into the area.

■ ***Concrete Real-World Example:*** *Disabling summary LSA acceptance on Stub Area 0.0.0.2 so branch routers have only intra-area routes plus a default route.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.2 type stub accepts-summary-lsa no
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Click Area 0.0.0.2.
2. Type: stub → Uncheck **Accept Summary LSA** → **OK** → **Commit**.

Topic 26: Advance-Routing: IPv4 OSPF Lab-Totally NSSA Area

■ **Meaning & Architectural Concept:** Not-So-Stubby Area that blocks external Type 5 LSAs from the backbone but permits local route redistribution via Type 7 LSAs. Totally NSSA also suppresses Type 3 summary LSAs.

■ ***Concrete Real-World Example:*** *Configuring Area 0.0.0.3 as Totally NSSA so a local router can redistribute RIP/BGP routes while keeping the area shielded from outside LSAs.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.3 type nssa accepts-summary-lsa no
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Add Area 0.0.0.3.
2. Type: nssa → Uncheck **Accept Summary LSA** → **OK** → **Commit**.

Topic 27: Advance-Routing: IPv4 OSPF Lab-External Summarization

■ **Meaning & Architectural Concept:** Summarizing external routes redistributed into OSPF at the ASBR boundary. A single Type 5 summary LSA is flooded instead of dozens of granular external subnets.

■ ***Concrete Real-World Example:*** *Summarizing redistributed internal server subnets under 10.0.0.0/8 so neighboring OSPF routers receive one aggregate route.*

```
admin@PA-VM# set network virtual-router default protocol ospf asbr-summary 10.0.0.0/8 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → **ASBR Summary**.
2. Click **Add** → Prefix: 10.0.0.0/8 → Check **Enable** → **OK** → **Commit**.

Topic 28: Advance-Routing: IPv4 OSPF Lab-Inter-Area Summarization

■ **Meaning & Architectural Concept:** Summarizing multiple contiguous /24 prefixes within an area at the Area Border Router (ABR) boundary into a single Type 3 Summary LSA advertised into Area 0.

■ ***Concrete Real-World Example:*** *Aggregating subnets 192.168.1.0/24 through 192.168.4.0/24 into 192.168.0.0/16 at the Area 1 border.*

```
admin@PA-VM# set network virtual-router default protocol ospf area 0.0.0.1 range 192.168.0.0/16 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → Click Area 0.0.0.1.
2. Select **Range** → Add Range: 192.168.0.0/16 → **OK** → **Commit**.

Topic 29: Advance-Routing: IPv4 OSPF Lab-Default-Information-Originate

■ **Meaning & Architectural Concept:** Generating a Type 5 External default route (0.0.0.0/0) and advertising it across all OSPF areas so all internal routers use the firewall as their egress Internet gateway.

■ ***Concrete Real-World Example:*** *Originating default route with metric 10 and metric type ext-2 into the OSPF domain.*

```
admin@PA-VM# set network virtual-router default protocol ospf default-information-originate enable yes metric 10 metric-type ext-2
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPF** → **Export Rules**.
2. Check **Default Information Originate** → Metric: 10, Metric Type: ext-2 → **OK** → **Commit**.

Topic 30: Advance-Routing: IPv4 OSPF Lab-Redistribution

■ **Meaning & Architectural Concept:** Exporting Connected, Static, or BGP routes into the OSPF domain via redistribution profiles, converting foreign routes into OSPF Type 5 External LSAs.

■ ***Concrete Real-World Example:*** *Redistributing all directly connected interfaces into OSPF with metric 20 and metric-type ext-2.*

```
admin@PA-VM# set network virtual-router default protocol ospf export-rules Redist-LAN enable yes metric 20 metric-type ext-2
admin@PA-VM# set network virtual-router default redist-profile Redist-Profile-1 filter match-route-type connect
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **Redistribution Profiles** → Add Redist-Profile-1 (Filter: connect).
2. Go to **OSPF** → **Export Rules** → Add rule referencing profile → **Commit**.

Topic 31: Advance-Routing: IPv4 OSPF Troubleshoot

■ **Meaning & Architectural Concept:** Diagnosing OSPF neighbor stalls: stuck in Init (one-way hello/security policy dropping multicast 224.0.0.5), stuck in 2-Way (DROther behavior), or stuck in ExStart/Exchange (MTU mismatch).

■ ***Concrete Real-World Example:*** *Tailing routed.log to detect MTU mismatch where firewall MTU is 1500 and peer MTU is 9000, causing DBD exchange drops.*

```
admin@PA-VM> show routing protocol ospf status
admin@PA-VM> show routing protocol ospf interface
admin@PA-VM> debug routing pfe ospf all on
admin@PA-VM> tail follow yes mp-log routed.log
```

Web GUI Step-by-Step Navigation:
1. Go to: **Monitor** → **Logs** → **System** → Filter: (description contains 'ospf').
2. Verify interface MTU matches neighbor; confirm security policy allows IP protocol 89.

Module 5: Enterprise IPv4 BGP (Border Gateway Protocol) Architecture

Topic 32: Advance-Routing: IPv4 BGP GUI/CLI

■ **Meaning & Architectural Concept:** Enabling the path-vector Border Gateway Protocol (BGPv4) daemon on the firewall virtual router, configuring the local Autonomous System Number (ASN), and establishing the Router-ID.

■ ***Concrete Real-World Example:*** *Configuring local private AS 65001 with BGP Router-ID 1.1.1.1 on virtual router default.*

```
admin@PA-VM# set network virtual-router default protocol bgp enable yes as 65001 router-id 1.1.1.1
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Virtual Routers** → Click **default** → **BGP** tab.
2. Check **Enable** → AS Number: 65001 → Router ID: 1.1.1.1 → **OK** → **Commit**.

Topic 33: Advance-Routing: IPv4 BGP Lab-Basic

■ **Meaning & Architectural Concept:** Configuring a Peer Group and establishing an external BGP (eBGP) peering session over TCP port 179 to an upstream ISP router in a different autonomous system.

■ ***Concrete Real-World Example:*** *Establishing eBGP session from local interface ethernet1/3 (192.168.3.2 in AS 65001) to ISP border router 192.168.3.1 in AS 65002.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-ISP type external
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-ISP peer ISP-Router peer-as 65002 local-address ip 192.168.3.
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Peer Group** → Click **Add**.
2. Name: eBGP-ISP → Type: ebgp.
3. Peer: Add ISP-Router → Peer AS: 65002, Peer IP: 192.168.3.1, Interface: ethernet1/3 → **OK** → **Commit**.

Topic 34: Advance-Routing: IPv4 BGP Lab-Neighbor Table Verification

■ **Meaning & Architectural Concept:** Auditing the BGP neighbor state machine. Verifying that the TCP 179 session has progressed through Idle, Connect, Active, OpenSent, OpenConfirm to Established.

■ ***Concrete Real-World Example:*** *Running show routing protocol bgp summary to confirm peer 192.168.3.1 is in Established state with active prefix exchange.*

```
admin@PA-VM> show routing protocol bgp peer
admin@PA-VM> show routing protocol bgp summary
```

Web GUI Step-by-Step Navigation:
1. Go to: **Virtual Routers** → **More Runtime Stats** → **BGP** → **Peer**.
2. Confirm Status = **Established**, Uptime is active, and prefixes are received.

Topic 35: Advance-Routing: IPv4 BGP Lab-BGP Table Verification

■ **Meaning & Architectural Concept:** Inspecting the BGP Link-State RIB table (Loc-RIB, Adj-RIB-In, Adj-RIB-Out). Auditing path attributes including AS-Path, Weight, Local Preference, Multi-Exit Discriminator (MED), and Next-Hop.

■ ***Concrete Real-World Example:*** *Verifying route 10.30.1.0/24 in Loc-RIB with AS-Path 65002 and Local Preference 100.*

```
admin@PA-VM> show routing protocol bgp loc-rib
admin@PA-VM> show routing protocol bgp rib-out
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **BGP** → **Loc-RIB**.
2. Inspect path attributes (AS-Path, Next Hop, Local Preference, MED) for all advertised and received routes.

Topic 36: Advance-Routing: IPv4 BGP Lab-Routing Table Verification

■ **Meaning & Architectural Concept:** Verifying that the best BGP routes selected by the BGP decision algorithm have been successfully committed to the active FIB with flag B (BGP Unicast).

■ ***Concrete Real-World Example:*** *Confirming external prefix 10.30.1.0/24 is active in the routing table with next-hop 192.168.3.1 and flag A S / A B.*

```
admin@PA-VM> show routing route protocol bgp virtual-router default
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **Routing Table**.
2. Filter by flag **B** to inspect all operational BGP forwarding entries.

Topic 37: Advance-Routing: IPv4 BGP Lab-Next-Hop-Self

■ **Meaning & Architectural Concept:** When an eBGP route is advertised to an internal iBGP peer, BGP preserves the original external next-hop IP by default. Next-Hop-Self forces the firewall to rewrite the next-hop IP to its own internal interface address so internal peers can resolve the route.

■ ***Concrete Real-World Example:*** *Configuring Next-Hop-Self on peer Core-Switch so that all eBGP ISP routes have next-hop rewritten to the firewall internal IP 192.168.1.2.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group iBGP-Peers peer Core-Switch next-hop use-self
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Peer Group** → Select peer.
2. In *Connection Options* → Next Hop: Select **Use Self** → **OK** → **Commit**.

Topic 38: Advance-Routing: IPv4 BGP Lab-Route-Reflector-Client

■ **Meaning & Architectural Concept:** Internal BGP (iBGP) prohibits advertising routes learned from one iBGP peer to another iBGP peer (split-horizon). Route Reflectors bypass this rule, allowing routers to reflect iBGP routes without requiring a full mesh of peerings.

■ ***Concrete Real-World Example:*** *Designating Core-Switch as a Route Reflector Client under cluster-id 1.1.1.1 on the firewall.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group iBGP-Peers peer Core-Switch ref-client client
admin@PA-VM# set network virtual-router default protocol bgp router-id 1.1.1.1 cluster-id 1.1.1.1
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select iBGP Peer.
2. Set **Reflector Client**: client → Set Cluster ID: 1.1.1.1 → **OK** → **Commit**.

Topic 39: Advance-Routing: IPv4 BGP Lab-Authentication

■ **Meaning & Architectural Concept:** Enforcing TCP MD5 cryptographic signatures (RFC 2385) on BGP TCP port 179 connections. Prevents TCP reset attacks and spoofed BGP route injections.

■ ***Concrete Real-World Example:*** *Configuring MD5 password BGPSecretKey99! on the eBGP ISP peering.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-ISP peer ISP-Router authentication md5 key BGPSecretKey99!
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select Peer → **Authentication**.
2. Check **Authentication** → Type: MD5 → Key: BGPSecretKey99! → **OK** → **Commit**.

Topic 40: Advance-Routing: IPv4 BGP Lab-Aggregation

■ **Meaning & Architectural Concept:** Generating a summarized CIDR supernet advertisement while suppressing granular internal subnets (summary-only). Conserves global BGP routing table memory.

■ ***Concrete Real-World Example:*** *Aggregating subnets under 10.0.0.0/8 and advertising only the /8 supernet to the ISP while suppressing internal /24 subnets.*

```
admin@PA-VM# set network virtual-router default protocol bgp aggregate 10.0.0.0/8 enable yes summary-only yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Aggregations**.
2. Click **Add** → Prefix: 10.0.0.0/8 → Check **Enable** → Check **Summary Only** → **OK** → **Commit**.

Topic 41: Advance-Routing: IPv4 BGP Lab-Redistribution

■ **Meaning & Architectural Concept:** Injecting connected, static, or OSPF routes into BGP advertisements sent to external autonomous systems via export rules and redistribution profiles.

■ ***Concrete Real-World Example:*** *Exporting corporate public IP block 192.168.1.0/24 into BGP to announce it to the upstream ISP.*

```
admin@PA-VM# set network virtual-router default protocol bgp export-rules Export-to-ISP enable yes match-address-prefix 192.168.1.0/24
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Export** tab.
2. Click **Add** → Name: Export-to-ISP → Match Prefix: 192.168.1.0/24 → **OK** → **Commit**.

Topic 42: Advance-Routing: IPv4 BGP Troubleshoot

■ **Meaning & Architectural Concept:** Resolving BGP state machine failures. Diagnosing why a peer is stuck in Active state (TCP 179 dropped by firewall security policy, wrong peer IP, wrong local IP, or eBGP multihop TTL issue).

■ ***Concrete Real-World Example:*** *Resetting a stuck BGP peering session and inspecting routed.log for AS-Path loop detection or TCP handshake resets.*

```
admin@PA-VM> show routing protocol bgp summary
admin@PA-VM> show routing protocol bgp peer detail
admin@PA-VM> test routing bgp peer-reset peer-group eBGP-ISP
admin@PA-VM> tail follow yes mp-log routed.log
```

Web GUI Step-by-Step Navigation:
1. Go to: **Monitor** → **Logs** → **System** → Filter: (description contains 'bgp').
2. If peering is in 'Active' state, ensure Security Policy allows application 'bgp' (TCP 179) between the peering zones.

Module 6: Advanced IPv6 Routing, Default Gateways & Path Monitoring

Topic 43: Advance-Routing: IPv6 Static Routing GUI/CLI

■ **Meaning & Architectural Concept:** Configuring 128-bit IPv6 static route entries in the virtual router pointing to an IPv6 next-hop address or egress interface.

■ ***Concrete Real-World Example:*** *Routing IPv6 enterprise subnet 2001:db8:2::/64 out ethernet1/2 to next-hop 2001:db8:2::1.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route-ipv6 Route-IPv6-DMZ1 destination 2001:db8:2::/64 interface e
admin@PA-VM# commit
admin@PA-VM> show routing route ipv6-only yes
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **Static Routes** → **IPv6**.
2. Click **Add** → Dest: 2001:db8:2::/64, Interface: ethernet1/2, Next Hop: 2001:db8:2::1 → **OK** → **Commit**.

Topic 44: Advance-Routing: IPv6 Default Routing GUI/CLI

■ **Meaning & Architectural Concept:** Configuring the global IPv6 default route (::/0) forwarding all unmapped IPv6 internet traffic out the external WAN gateway.

■ ***Concrete Real-World Example:*** *Forwarding all IPv6 traffic out ethernet1/3 to the ISP IPv6 gateway at 2001:db8:3::1.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route-ipv6 Default-IPv6 destination ::/0 interface ethe net1/3 nex
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **Static Routes** → **IPv6**.
2. Click **Add** → Dest: ::/0 → Interface: ethernet1/3 → Next Hop: 2001:db8:3::1 → **OK** → **Commit**.

Topic 45: Advance-Routing: IPv6 Path Monitoring GUI/CLI

■ **Meaning & Architectural Concept:** Attaching ICMPv6 ping health checks to an IPv6 static route. If the monitored IPv6 address stops responding, the route is pulled from the FIB to prevent blackholing IPv6 traffic.

■ ***Concrete Real-World Example:*** *Monitoring Google Public IPv6 DNS 2001:4860:4860::8888 every 3s. If 5 pings fail, deactivate primary IPv6 default route.*

```
admin@PA-VM# set network virtual-router default routing-table ip static-route-ipv6 Default-IPv6 path-monitor enable yes failure condition
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **Static Routes** → **IPv6** → Edit Default-IPv6 → **Path Monitor**.
2. Check **Enable** → Target: 2001:4860:4860::8888, Interval: 3, Count: 5 → **OK** → **Commit**.

Module 7: Enterprise IPv6 OSPFv3 Architecture

Topic 46: Advance-Routing: IPv6 OSPFv3 via GUI/CLI

■ **Meaning & Architectural Concept:** Enabling the OSPFv3 protocol daemon designed specifically for IPv6. Note that OSPFv3 still requires a 32-bit dotted-decimal Router-ID (e.g. 1.1.1.1).

■ ***Concrete Real-World Example:*** *Enabling OSPFv3 on virtual router default with Router-ID 1.1.1.1.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 enable yes router-id 1.1.1.1
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. Navigate to: **Network** → **Virtual Routers** → default → **OSPFv3** tab.
2. Check **Enable** → Router ID: 1.1.1.1 → **OK** → **Commit**.

Topic 47: Advance-Routing: IPv6 OSPFv3 Lab-Basic and Multi-Area

■ **Meaning & Architectural Concept:** Configuring multi-area OSPFv3 topology. Links in Area 0 form the IPv6 backbone while Area 1 isolates branch networks.

■ ***Concrete Real-World Example:*** *Placing ethernet1/1 in OSPFv3 Area 0.0.0.0 and ethernet1/2 in Area 0.0.0.1.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.0 interface ethernet1/1 enable yes
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.1 interface ethernet1/2 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → **Areas**.
2. Add Area 0.0.0.0 (Interface: ethernet1/1) and Area 0.0.0.1 (Interface: ethernet1/2) → **Commit**.

Topic 48: Advance-Routing: IPv6 OSPFv3 Lab-Neighbor Table Verification

■ **Meaning & Architectural Concept:** Auditing OSPFv3 neighbor adjacencies. In OSPFv3, neighbors form adjacencies over IPv6 Link-Local addresses (fe80::) rather than global unicast addresses.

■ ***Concrete Real-World Example:*** *Verifying neighbor status shows Full with peer link-local address fe80::ce1:c3ff:fe5e:0.*

```
admin@PA-VM> show routing protocol ospfv3 neighbor
admin@PA-VM> show routing protocol ospfv3 neighbor detail
```

Web GUI Step-by-Step Navigation:
1. Go to: **Virtual Routers** → **More Runtime Stats** → **OSPFv3** → **Neighbors**.
2. Confirm neighbor status shows **Full** over IPv6 link-local (fe80::) addresses.

Topic 49: Advance-Routing: IPv6 OSPFv3 Lab-Database Table Verification

■ **Meaning & Architectural Concept:** Auditing the OSPFv3 Link-State Database (LSDB). Verifying Link LSAs (Type 8, flooding local link-local IPs) and Intra-Area-Prefix LSAs (Type 9, carrying IPv6 prefixes decoupled from topology).

■ ***Concrete Real-World Example:*** *Running show routing protocol ospfv3 lsdb to verify Type 8 and Type 9 LSA synchronization.*

```
admin@PA-VM> show routing protocol ospfv3 lsdb
admin@PA-VM> show routing protocol ospfv3 dump
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **OSPFv3** → **Link State Database**.
2. Audit Type 8 (Link) and Type 9 (Intra-Area-Prefix) LSA tables.

Topic 50: Advance-Routing: IPv6 OSPFv3 Lab-Routing Table Verification

■ **Meaning & Architectural Concept:** Confirming computed OSPFv3 shortest paths installed into the active IPv6 forwarding table (FIB).

■ ***Concrete Real-World Example:*** *Verifying learned IPv6 subnet 2001:db8:10::/64 installed with next-hop fe80::1 via ethernet1/1.*

```
admin@PA-VM> show routing protocol ospfv3 route
admin@PA-VM> show routing route protocol ospfv3 virtual-router default
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **Routing Table**.
2. Filter for IPv6 routes learned via OSPFv3 to verify forwarding metrics.

Topic 51: Advance-Routing: IPv6 OSPFv3 Lab-Virtual-Links

■ **Meaning & Architectural Concept:** Bridging a disconnected or partitioned IPv6 OSPFv3 area back to Area 0 across a transit area.

■ ***Concrete Real-World Example:*** *Creating virtual link VL-IPv6-R3 across transit Area 1 to reach isolated router 3.3.3.3.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.1 virtual-link VL-IPv6-R3 neighbor-id 3.3.3.3 transit-area-id 0
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Area 0.0.0.1.
2. Select **Virtual Link** tab → Add VL-IPv6-R3 → Neighbor: 3.3.3.3 → **OK** → **Commit**.

Topic 52: Advance-Routing: IPv6 OSPFv3 Lab-Hello Interval Modification

■ **Meaning & Architectural Concept:** Modifying the OSPFv3 Hello timer on an IPv6 interface to speed up link failure detection.

■ ***Concrete Real-World Example:*** *Lowering Hello timer to 5s on interface ethernet1/1 in Area 0.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.0 interface ethernet1/1 hello-interval 5
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Area 0.0.0.0 → Click interface.
2. Set **Hello Interval**: 5 → **OK** → **Commit**.

Topic 53: Advance-Routing: IPv6 OSPFv3 Lab-Dead Interval Modification

■ **Meaning & Architectural Concept:** Adjusting the OSPFv3 Dead timer (must match across peers) before dropping an unresponsive neighbor.

■ ***Concrete Real-World Example:*** *Setting Dead timer to 20s (4x Hello timer of 5s) on ethernet1/1.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.0 interface ethernet1/1 dead-counts 20
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Area 0.0.0.0 → Click interface.
2. Set **Dead Interval**: 20 → **OK** → **Commit**.

Topic 54: Advance-Routing: IPv6 OSPFv3 Lab-Stub Area

■ **Meaning & Architectural Concept:** Configuring an IPv6 OSPF area to block external Type 5 LSAs, auto-injecting an IPv6 default route (::/0).

■ ***Concrete Real-World Example:*** *Configuring branch Area 0.0.0.2 as Stub with default route metric 10.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.2 type stub default-route metric 10
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Add Area 0.0.0.2.
2. Type: stub → Default Route Metric: 10 → **OK** → **Commit**.

Topic 55: Advance-Routing: IPv6 OSPFv3 Lab-Totally Stub Area

■ **Meaning & Architectural Concept:** Blocking both External LSAs and Inter-Area Summary LSAs in IPv6, leaving only intra-area routes and a default route ::/0.

■ ***Concrete Real-World Example:*** *Unchecking Accept Summary LSA on Area 0.0.0.2 to optimize branch memory.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.2 type stub accepts-summary-lsa no
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Area 0.0.0.2.
2. Type: stub → Uncheck **Accept Summary LSA** → **OK** → **Commit**.

Topic 56: Advance-Routing: IPv6 OSPFv3 Lab-Totally NSSA Area

■ **Meaning & Architectural Concept:** Permitting local IPv6 ASBR redistribution via Type 7 LSAs while blocking external Type 5 and summary Type 3 LSAs.

■ ***Concrete Real-World Example:*** *Configuring Area 0.0.0.3 as Totally NSSA for a local dual-stack router.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.3 type nssa accepts-summary-lsa no
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Add Area 0.0.0.3.
2. Type: nssa → Uncheck **Accept Summary LSA** → **OK** → **Commit**.

Topic 57: Advance-Routing: IPv6 OSPFv3 Lab-External Summarization

■ **Meaning & Architectural Concept:** Summarizing redistributed external IPv6 routes at the ASBR boundary into a single aggregate LSA.

■ ***Concrete Real-World Example:*** *Summarizing external IPv6 subnets under 2001:db8::/32 at the firewall ASBR boundary.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 asbr-summary 2001:db8::/32 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → **ASBR Summary**.
2. Add Prefix: 2001:db8::/32 → **OK** → **Commit**.

Topic 58: Advance-Routing: IPv6 OSPFv3 Lab-Inter-Area Summarization

■ **Meaning & Architectural Concept:** Summarizing internal IPv6 subnets at the ABR boundary into a single Type 3 Summary LSA advertised into Area 0.

■ ***Concrete Real-World Example:*** *Aggregating branch /64 subnets under 2001:db8:1::/48 at the Area 1 border.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 area 0.0.0.1 range 2001:db8:1::/48 enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → Area 0.0.0.1.
2. Range tab → Add: 2001:db8:1::/48 → **OK** → **Commit**.

Topic 59: Advance-Routing: IPv6 OSPFv3 Lab-Default-Information-Originate

■ **Meaning & Architectural Concept:** Originating an IPv6 default route (::/0) into OSPFv3 so internal IPv6 nodes route outbound traffic through the firewall.

■ ***Concrete Real-World Example:*** *Advertising default route ::/0 with metric 10 and metric-type ext-2 into OSPFv3.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 default-information-originate enable yes metric 10 metric-type ext-2
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → **Export Rules**.
2. Check **Default Information Originate** → Metric: 10 → **OK** → **Commit**.

Topic 60: Advance-Routing: IPv6 OSPFv3 Lab-Redistribution

■ **Meaning & Architectural Concept:** Injecting connected or static IPv6 routes into OSPFv3 via export rules.

■ ***Concrete Real-World Example:*** *Redistributing local connected IPv6 subnets into OSPFv3 with metric 20.*

```
admin@PA-VM# set network virtual-router default protocol ospfv3 export-rules Redist-IPv6 enable yes metric 20
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **OSPFv3** → **Export Rules**.
2. Add export rule referencing IPv6 redistribution profile → **Commit**.

Topic 61: Advance-Routing: IPv6 OSPFv3 Troubleshoot

■ **Meaning & Architectural Concept:** Diagnosing OSPFv3 issues including missing link-local addresses, IP protocol 89 drops, and MTU mismatches.

■ ***Concrete Real-World Example:*** *Tailing routed.log to detect why an OSPFv3 neighbor is stuck in Init due to link-local fe80:: address unreachable.*

```
admin@PA-VM> show routing protocol ospfv3 status
admin@PA-VM> show routing protocol ospfv3 interface
admin@PA-VM> debug routing pfe ospfv3 all on
admin@PA-VM> tail follow yes mp-log routed.log
```

Web GUI Step-by-Step Navigation:
1. Go to: **Monitor** → **Logs** → **System** → Filter: (description contains 'ospfv3').
2. Verify IPv6 link-local addressing on interfaces and check for security policy drops on protocol 89.

Module 8: Enterprise IPv6 Multiprotocol BGP (MP-BGP / BGPv6) Architecture

Topic 62: Advance-Routing: IPv6 BGPv6 via GUI/CLI

■ **Meaning & Architectural Concept:** Enabling the Multiprotocol BGP (MP-BGP) extension for IPv6 Address Family Identifier (AFI IPv6). Allows exchange of IPv6 Network Layer Reachability Information (NLRI).

■ ***Concrete Real-World Example:*** *Activating BGP on virtual router default with local AS 65001 and configuring an external IPv6 peer in AS 65002.*

```
admin@PA-VM# set network virtual-router default protocol bgp enable yes as 65001 router-id 1.1.1.1
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-IPv6-ISP type external
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-IPv6-ISP peer ISP-IPv6-Peer peer-as 65002 local-address ip 20
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Peer Group** → Add eBGP-IPv6-ISP.
2. Add Peer → Peer AS: 65002, Peer IP: 2001:db8:3::1, Address Family: **IPv6** → **OK** → **Commit**.

Topic 63: Advance-Routing: IPv6 BGPv6 Lab-Basic

■ **Meaning & Architectural Concept:** Establishing and verifying an external IPv6 BGP peering session over TCP port 179.

■ ***Concrete Real-World Example:*** *Activating the IPv6 BGP session to upstream telecom router 2001:db8:3::1.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-IPv6-ISP peer ISP-IPv6-Peer enable yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select ISP-IPv6-Peer.
2. Ensure **Enable** is checked → **OK** → **Commit**.

Topic 64: Advance-Routing: IPv6 BGPv6 Lab-Neighbor Table Verification

■ **Meaning & Architectural Concept:** Auditing the IPv6 BGP neighbor state to verify the session has reached Established status for Address Family IPv6.

■ ***Concrete Real-World Example:*** *Checking show routing protocol bgp peer afi ipv6 to confirm 2001:db8:3::1 is Established.*

```
admin@PA-VM> show routing protocol bgp peer afi ipv6
admin@PA-VM> show routing protocol bgp summary
```

Web GUI Step-by-Step Navigation:
1. Go to: **Virtual Routers** → **More Runtime Stats** → **BGP** → **Peer**.
2. Confirm IPv6 neighbor shows Status = **Established**.

Topic 65: Advance-Routing: IPv6 BGPv6 Lab-BGP Table Verification

■ **Meaning & Architectural Concept:** Inspecting the IPv6 Loc-RIB and Adj-RIB tables to evaluate received IPv6 prefixes, AS-Path, and next-hop reachability.

■ ***Concrete Real-World Example:*** *Verifying learned prefix 2001:db8:30::/64 in Loc-RIB with AS-Path 65002.*

```
admin@PA-VM> show routing protocol bgp loc-rib afi ipv6
admin@PA-VM> show routing protocol bgp rib-out afi ipv6
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **BGP** → **Loc-RIB**.
2. Select **IPv6** address family to inspect path attributes.

Topic 66: Advance-Routing: IPv6 BGPv6 Lab-Routing Table Verification

■ **Meaning & Architectural Concept:** Verifying best BGPv6 routes committed into the active IPv6 forwarding table with flag B.

■ ***Concrete Real-World Example:*** *Confirming active IPv6 BGP route 2001:db8:30::/64 with next-hop 2001:db8:3::1 installed in the FIB.*

```
admin@PA-VM> show routing route protocol bgp ipv6-only yes virtual-router default
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → **More Runtime Stats** → **Routing Table**.
2. Filter for IPv6 and flag **B** to inspect committed next-hops.

Topic 67: Advance-Routing: IPv6 BGPv6 Lab-Next-Hop-Self

■ **Meaning & Architectural Concept:** Rewriting external IPv6 next-hop addresses to the firewall local IPv6 interface when advertising eBGP routes to internal iBGPv6 peers.

■ ***Concrete Real-World Example:*** *Configuring Next-Hop-Self for internal peer Internal-Core-v6 so external IPv6 routes resolve to the firewall internal IPv6 address.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group iBGP-IPv6 peer Internal-Core-v6 next-hop use-self
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select iBGPv6 peer.
2. In *Connection Options* → Next Hop: Select **Use Self** → **OK** → **Commit**.

Topic 68: Advance-Routing: IPv6 BGPv6 Lab-Route-Reflector-Client

■ **Meaning & Architectural Concept:** Reflecting iBGPv6 routes between internal peers to eliminate the need for a full-mesh topology in large enterprise IPv6 networks.

■ ***Concrete Real-World Example:*** *Designating Internal-Core-v6 as a Route Reflector Client under cluster-id 1.1.1.1.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group iBGP-IPv6 peer Internal-Core-v6 ref-client client
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select iBGPv6 peer.
2. Reflector Client: Select **client** → **OK** → **Commit**.

Topic 69: Advance-Routing: IPv6 BGPv6 Lab-Authentication

■ **Meaning & Architectural Concept:** Securing IPv6 BGP peering sessions using MD5 TCP checksums to block unauthorized session hijacking or spoofing.

■ ***Concrete Real-World Example:*** *Configuring MD5 password IPv6BgpPass123! on IPv6 peering to ISP.*

```
admin@PA-VM# set network virtual-router default protocol bgp peer-group eBGP-IPv6-ISP peer ISP-IPv6-Peer authentication md5 key IPv6BgpPas
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → Select IPv6 peer → **Authentication**.
2. Type: MD5 → Key: IPv6BgpPass123! → **OK** → **Commit**.

Topic 70: Advance-Routing: IPv6 BGPv6 Lab-Aggregation

■ **Meaning & Architectural Concept:** Aggregating granular internal IPv6 subnets into a single /32 or /48 supernet advertisement to global transit providers while suppressing internal subnets.

■ ***Concrete Real-World Example:*** *Aggregating enterprise IPv6 allocation under 2001:db8::/32 with summary-only enabled.*

```
admin@PA-VM# set network virtual-router default protocol bgp aggregate-ipv6 2001:db8::/32 enable yes summary-only yes
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Aggregations (IPv6)**.
2. Click **Add** → Prefix: 2001:db8::/32 → Check **Summary Only** → **OK** → **Commit**.

Topic 71: Advance-Routing: IPv6 BGPv6 Lab-Redistribution

■ **Meaning & Architectural Concept:** Exporting corporate IPv6 prefixes (connected, static, or OSPFv3) into BGPv6 updates sent to upstream ISP transit routers.

■ ***Concrete Real-World Example:*** *Exporting enterprise public prefix 2001:db8:1::/48 into BGPv6.*

```
admin@PA-VM# set network virtual-router default protocol bgp export-rules Export-IPv6-BGP enable yes match-address-prefix 2001:db8:1::/48
admin@PA-VM# commit
```

Web GUI Step-by-Step Navigation:
1. In **Virtual Routers** → default → **BGP** → **Export (IPv6)**.
2. Add export rule matching enterprise prefix 2001:db8:1::/48 → **OK** → **Commit**.

Topic 72: Advance-Routing: IPv6 BGPv6 Troubleshoot

■ **Meaning & Architectural Concept:** Diagnosing IPv6 BGP peering failures: verifying TCP 179 connectivity over IPv6, ensuring security policy permits application bgp, resolving Address Family negotiation mismatches, and inspecting next-hop reachability.

■ ***Concrete Real-World Example:*** *Tailing routed.log to detect AFI negotiation failures or link-local next-hop resolution issues with the upstream ISP.*

```
admin@PA-VM> show routing protocol bgp peer afi ipv6 detail
admin@PA-VM> show routing protocol bgp loc-rib afi ipv6
admin@PA-VM> debug routing pfe bgp all on
admin@PA-VM> tail follow yes mp-log routed.log
```

Web GUI Step-by-Step Navigation:
1. Go to: **Monitor** → **Logs** → **System** → Filter: (subtype eq routing) and (description contains 'bgp').
2. Confirm IPv6 TCP 179 reachability; verify security policy allows application 'bgp' between peering zones.

ENGINEERING STANDARDS & CURRICULUM COMPLIANCE

Architectural Certification Sign-Off: This manual covers all 72 routing modules required for production data center deployments and Palo Alto Networks PCNSA / PCNSE professional certifications. Every command, technical concept, and GUI navigation path adheres to the PAN-OS Virtual Router standards across standalone, High Availability, and multi-tenant Virtual System (vsys) topologies.